

B.A./B.Sc. (Hons.) Semester - V Examination 2024-25
STATISTICS

Paper No: STA/STB-502: Statistical Inference and Decision Theory

Time: 3 hours

Full Marks: 70

Note: Attempt any five questions in all. All questions carry equal marks.

1. (a) Define the terms Estimator, estimate, and standard error[cite: 5656]. Let $X_1, X_2, \dots, X_n$ be a sample of size n from a distribution with unknown mean μ ($-\infty < \mu < \infty$) and unknown variance $\sigma^2 > 0$ [cite: 5656]. Show that the sample mean statistic $\bar{X}$ and the weighted statistic:

$$Y = \frac{X_1 + 2X_2 + \dots + nX_n}{\frac{n(n+1)}{2}}$$

are both unbiased estimators of μ [cite: 5656]. Furthermore, establish the exact algebraic relation between $\text{Var}(\bar{X})$ and $\text{Var}(Y)$ [cite: 5657].

- (b) If T_1 is the minimum variance unbiased estimator (MVUE) of $\gamma(\theta)$, $\theta \in \Theta$, and T_2 is any other unbiased estimator of $\gamma(\theta)$ with efficiency e , prove that the correlation coefficient ρ between T_1 and T_2 is strictly given by $\rho = \sqrt{e}$ [cite: 5658].
2. (a) Define the property of consistency for an estimator[cite: 5659]. State and prove the standard sufficient conditions for an estimator to be consistent[cite: 5659].
- (b) Let $X_1, X_2, \dots, X_n$ be a random sample from the probability density function:

$$f(x|\theta) = \begin{cases} e^{-(x-\theta)} & ; x \geq \theta \\ 0 & ; \text{otherwise} \end{cases}$$

[cite: 5660] Show that the first order statistic $T_n = \text{Min}(X_1, X_2, \dots, X_n) = X_{(1)}$ is a consistent estimator of θ [cite: 5661].

3. (a) Define a sufficient statistic and a complete statistic[cite: 5662]. Let $X_1, X_2, \dots, X_n$ be a random sample from the probability density function:

$$f(x|\theta) = \begin{cases} \frac{1}{\theta} & ; 0 < x < \theta \\ 0 & ; \text{otherwise} \end{cases}$$

[cite: 5663, 5664, 5665] where $\theta \in (0, \infty)$ [cite: 5666].

Show that the largest order statistic $T = \text{Max}(X_1, X_2, \dots, X_n) = X_{(n)}$ is a complete sufficient statistic[cite: 5666].

- (b) Define the Minimum Variance Unbiased Estimator (MVUE)[cite: 5667]. Let $X_1, X_2, \dots, X_n$ be a random sample from a Poisson distribution $P(\theta)$, $\theta > 0$ [cite: 5667]. Let $\gamma(\theta) = e^{-\theta} = P[X = 0]$ [cite: 5668]. Find the explicit minimum variance unbiased estimator of $\gamma(\theta)$ [cite: 5668].
4. (a) Explain the method of maximum likelihood estimation[cite: 5669]. Let $X_1, X_2, \dots, X_n$ be a random sample from a Normal distribution where both the mean and the variance are equal to θ , i.e., $N(\theta, \theta)$ [cite: 5669]. Obtain the maximum likelihood estimator (MLE) of θ [cite: 5669].
- (b) Explain the method of moments for the estimation of a parameter[cite: 5672]. Suppose $X_1, X_2, \dots, X_n$ is a random sample from a continuous uniform population with density function:

$$f(x|\alpha, \beta) = \begin{cases} \frac{1}{\beta-\alpha} & ; \alpha < x < \beta \\ 0 & ; \text{otherwise} \end{cases}$$

[cite: 5673, 5674] Find the estimators of α and β by the method of moments[cite: 5675].

5. What is a confidence interval, and how does it differ fundamentally from a point estimate[cite: 5676]? Define the terms confidence limits and length of a confidence interval[cite: 5677]. Let $X_1, X_2, \dots, X_n$ be a random sample from a Uniform distribution $U(0, \theta)$ [cite: 5678]. Show that $Q = \frac{X_{(n)}}{\theta}$ is a valid pivotal quantity[cite: 5678]. Utilizing this pivotal quantity, derive a $100(1 - \alpha)\%$ confidence interval for the parameter θ [cite: 5679].
6. (a) State and prove the Neyman-Pearson Lemma[cite: 5680].
 (b) Obtain the most powerful test for testing the null hypothesis $H_0 : \theta = \theta_0$ against the alternative hypothesis $H_1 : \theta = \theta_1$ on the basis of a random sample of size n from an exponential distribution with probability density function:

$$f(x|\theta) = \begin{cases} \theta e^{-x\theta} & ; x > 0, \theta > 0 \\ 0 & ; \text{otherwise} \end{cases}$$

[cite: 5681, 5682]

7. Describe the operational procedure of the Likelihood Ratio Test (LRT)[cite: 5683]. Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with unknown mean μ and unknown variance σ^2 [cite: 5683]. Obtain the likelihood ratio test of size α for testing the null hypothesis $H_0 : \sigma^2 = \sigma_0^2$ against the alternative hypothesis $H_1 : \sigma^2 \neq \sigma_0^2$ [cite: 5683].
8. Define loss function, Risk function, and the Minimax Rule in decision theory[cite: 5684]. Obtain the explicit Bayes estimator of the parameter θ under the Squared Error Loss Function (SELF) on the basis of a random sample of size n , where the prior distribution for θ follows a Gamma distribution $G(a, b)$ [cite: 5684].

External Portals & Academic Resources

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app/>